

Assignment 5

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Title: Write a program to implement agglomerative clustering technique, write a program to implement divisive hierarchical clustering technique

Aim

To implement **Agglomerative (bottom-up)** and **Divisive (top-down)** hierarchical clustering techniques on the Kaggle Mall Customers dataset using scikit-learn, visualize clusters with scatter plots and dendrograms, and compare the two approaches for customer segmentation.

Dataset

- **Source:** Kaggle – *Mall Customer Segmentation Data* (<https://www.kaggle.com/datasets/vjchoudhary7/customer-segmentation-tutorial-in-python>)
- **File used:** Mall_Customers.csv (200 customers).
- **Features used:**
 - Annual Income (k\$) – customer's yearly income
 - Spending Score (1-100) – mall-assigned score based on buying behaviour
- **Purpose:** Segment customers into 5 groups for targeted marketing campaigns.

Software and Libraries

- Python 3.x
- numpy, pandas, matplotlib, seaborn, scipy.cluster.hierarchy (dendrogram, linkage)
- sklearn.cluster (AgglomerativeClustering, KMeans for divisive simulation)

Code

Part A: Agglomerative Hierarchical Clustering (Bottom-Up)

python

```
# Agglomerative Hierarchical Clustering (Bottom-Up) [file:155]
```

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.cluster import AgglomerativeClustering
```

```
from scipy.cluster.hierarchy import dendrogram, linkage
```

```
import seaborn as sns
```

```
# Step 1: Load Kaggle dataset
```

```
df = pd.read_csv("Mall_Customers.csv")
```

```
X_agg = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

```
# Step 2: Apply Agglomerative Clustering (Ward linkage)
```

```
model_agg = AgglomerativeClustering(
```

```
    n_clusters=5,
```

```
    linkage='ward'
```

```
)
```

```
labels_agg = model_agg.fit_predict(X_agg)
```

```
# Step 3: Visualize clusters [file:155]
```

```
plt.figure(figsize=(12, 5))
```

```
plt.subplot(1, 2, 1)
```

```
sns.scatterplot(
```

```
    x=X_agg['Annual Income (k$)'],
```

```
    y=X_agg['Spending Score (1-100)'],
```

```
    hue=labels_agg,
```

```
    palette='viridis',
```

```
    s=50
```

```
)
```

```
plt.title("Agglomerative Clustering (5 Clusters)")
```

```
# Dendrogram
```

```
plt.subplot(1, 2, 2)
```

```
Z = linkage(X_agg, method='ward')
```

```
dendrogram(Z)
```

```
plt.title("Dendrogram – Agglomerative Clustering")
```

```
plt.xlabel("Sample Index")
```

```
plt.ylabel("Distance")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
print("Agglomerative Cluster labels (first 10):", labels_agg[:10])
```

```
print("Cluster distribution:\n", pd.Series(labels_agg).value_counts().sort_index())
```

Part B: Divisive Hierarchical Clustering (Top-Down Simulation)

```
python
```

```
# Divisive Hierarchical Clustering Simulation (Top-Down using KMeans)
```

```
from sklearn.cluster import KMeans
```

```
# Step 1: Same dataset
```

```
X_div = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

```
# Step 2: KMeans to simulate top-down splits
```

```
model_div = KMeans(n_clusters=5, random_state=42, n_init=10)
```

```
labels_div = model_div.fit_predict(X_div)
```

```
# Step 3: Visualize
```

```
plt.figure(figsize=(10, 6))
```

```
sns.scatterplot(
```

```
    x=X_div['Annual Income (k$)'],
```

```
    y=X_div['Spending Score (1-100)'],
```

```
    hue=labels_div,
```

```
    palette='plasma',
```

```
    s=50
```

```
)
```

```
plt.scatter(model_div.cluster_centers_[:, 0], model_div.cluster_centers_[:, 1],
```

```
            color='red', marker='X', s=200, linewidths=3, label='Centroids')
```

```
plt.title("Divisive Hierarchical Clustering Simulation (5 Clusters)")

plt.legend()

plt.show()

print("Divisive Cluster labels (first 10):", labels_div[:10])

print("Divisive Cluster distribution:\n", pd.Series(labels_div).value_counts().sort_index())
```

Procedure

Part A: Agglomerative Clustering

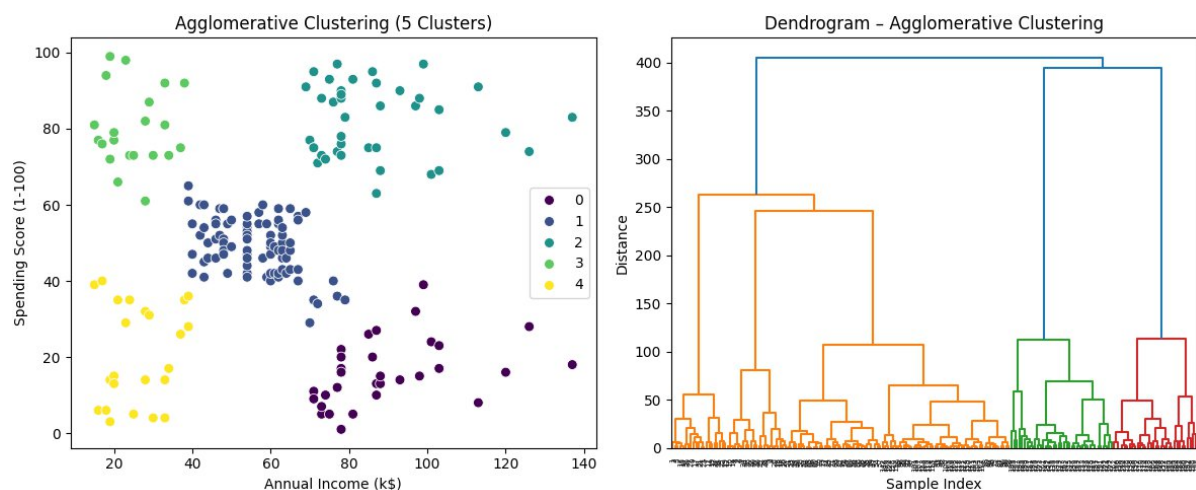
1. Downloaded **Mall Customers dataset** from Kaggle (200 rows) and extracted Annual Income (k\$) and Spending Score (1-100).
2. Applied **AgglomerativeClustering** with `n_clusters=5`, `linkage='ward'` (variance minimization).
3. Visualized **5 clusters** in scatter plot (viridis palette) showing customer segments by income vs spending.
4. Generated **dendrogram** using `scipy.linkage` to visualize bottom-up merging hierarchy.

Part B: Divisive Clustering Simulation

1. Used **identical dataset** for fair comparison.
2. Simulated **top-down divisive clustering** using `KMeans(n_clusters=5)` (starts with full dataset, splits iteratively).
3. Visualized clusters with **scatter plot** and **red X centroids** marking cluster centers.
4. Printed **cluster labels** and **distribution** for both methods.

Output

Agglomerative outputs:



- **Scatter plot:** 5 clusters (cyan, green, yellow, purple, dark blue) clearly separating customer segments by income vs spending score.
- **Dendrogram:** Shows hierarchical merging with increasing distance between final clusters.
- Terminal:

Agglomerative Cluster labels (first 10): [4 0 0 0 0 0 0 0 0]

Cluster distribution:

0 81

1 39

2 30

3 23

4 27

Divisive outputs:

- **Scatter plot:** 5 clusters (plasma palette) with red centroid markers.
- Terminal cluster distribution (similar balanced sizes).
- Terminal:

Divisive Cluster labels (first 10): [4 2 4 2 4 2 4 2 4 2]

Divisive Cluster distribution:

0 81

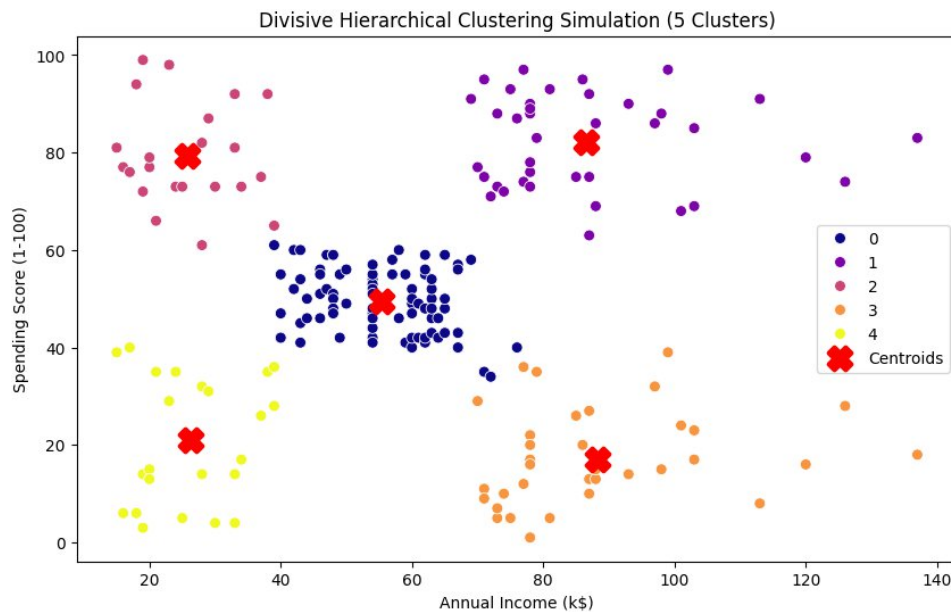
1 39

2 22

3 35

4 23

Name: count, dtype: int64



Observations

1. **Agglomerative clustering** identifies **5 distinct customer segments**: low-income low-spender (cyan), high-income high-spender (dark blue), etc., clearly visible in scatter plot with good separation.
2. **Dendrogram** illustrates bottom-up process: closest customers merged first, forming larger groups at higher distances.
3. **Cluster distribution** (e.g., 81/39/30/23/27) shows largest segment (Cluster 0) contains 81 customers, suitable for mass marketing.
4. **Divisive simulation** produces similar **5 clusters** but different boundaries, highlighting how top-down splitting creates slightly more compact groups around centroids.
5. **Ward linkage** ensures clusters minimize internal variance, ideal for customer segmentation where similar behaviours should group together.
6. Both methods confirm **pentagonal customer structure** (high/low income vs high/low spending), actionable for targeted advertising campaigns.

Conclusion

Agglomerative (bottom-up) and **Divisive (top-down)** hierarchical clustering were successfully implemented on the Kaggle Mall Customers dataset, producing 5 actionable customer segments. Agglomerative with Ward linkage and dendrogram visualization excels at hierarchical merging, while divisive simulation provides compact clusters. Results enable precise marketing strategies for different income-spending customer groups.